ATOM-SG Exam-Quality Baseline Test

P6 Mathematics | 40 Questions | 1 Hour | No Calculator

Name: _____ **Class:** _____ **Date:** _____

Instructions: Show all working. Exam-quality problems with professional diagrams.

Q 1. [H] [WP] Cross-Thread

Mrs. Lee spent $240 on groceries. She spent 2/5 of the remainder on a dress, then 1/3 of the new remainder on shoes. If she had $72 left, how much did she have at first?

Working:

Answer: _____ **(4 marks)**

Q 2. [H] [WP] Cross-Thread

A tank was 5/6 full. After pouring out 15 litres, it was 2/3 full. What is the capacity of the tank?

Working:

Answer: _____ **(3 marks)**

Q 3. [H] [WP] Cross-Thread

The ratio of red to blue balls was 3:5. After removing 12 red balls, the ratio became 1:3. How many red balls were there at first?

Working:

Answer: _____ **(4 marks)**

Q 4. [H] [WP] Cross-Thread

Alan and Ben's ages are in ratio 4:7. In 6 years, the ratio will be 2:3. How old is Alan now?

Working:

Answer: _____ **(4 marks)**

Q 5. [H] [WP] Cross-Thread

A shop had 240 pens. On Monday, they sold 2/5 of the pens. On Tuesday, they sold 1/3 of the remainder. On Wednesday, they sold 1/2 of the new remainder. How many pens were left?

Working:

Answer: _____ **(4 marks)**

Q 6. [H] [WP] Cross-Thread

Alice and Betty had $280. After Alice spent 1/3 of her money and Betty spent $40, they had equal amounts. How much did Alice have at first?

Working:

Answer: _____ **(4 marks)**

Q 7. [H] [WP] Cross-Thread

A baker made 360 cupcakes. He sold 1/4 in the morning and 2/3 of the remainder in the afternoon. How many cupcakes did he have left?

Working:

Answer: _____ **(3 marks)**

Q 8. [H] [WP] Cross-Thread

John has 3/5 as many stamps as Mary. Mary has 5/6 as many as Tom. If Tom has 72 stamps, how many does John have?

Working:

Answer: _____ **(3 marks)**

Q 9. [M] [WP] Part-Whole

The ratio of boys to girls in a class is 5:3. If there are 40 students altogether, how many boys are there?

Working:

Answer: _____ **(2 marks)**

Q10. [M] [WP] Before-After

A bus had 60 passengers. 15 got off at the first stop, then 8 at the second. How many were left?

Working:

Answer: _____ **(2 marks)**

Q11. [H] [WP] Cross-Thread

Ali and Ben had $280 altogether. After Ali gave 1/4 of his money to Ben, they had the same amount. How much did Ali have at first?

Working:

Answer: _____ **(4 marks)**

Q12. [H] [WP] Cross-Thread

A tank was full. After 3/7 was poured out, then 2/5 of the remainder was used, 48 litres were left. How much was in the tank at first?

Working:

Answer: _____ **(4 marks)**

Q13. [M] [WP] Part-Whole

There are 80 students. 3/8 are girls. How many boys are there?

Working:

Answer: _____ **(2 marks)**

Baseline Test - Continued

Q14. [H] [WP] Cross-Thread

A seller had 240 apples. He sold 3/4 to Shop A and 1/2 of the remainder to Shop B. How many were left?

Working:

Answer: _____ **(3 marks)**

Q15. [M] [WP] Part-Whole

Jenny and Kelly have 84 stickers. Jenny has 16 more than Kelly. How many does Kelly have?

Working:

Answer: _____ **(2 marks)**

Q16. [H] [WP] Cross-Thread

A bag's price increased by 20%, then had a 15% discount. If the final price was $204, what was the original?

Working:

Answer: _____ **(4 marks)**

Q17. [H] [WP] Part-Whole

Apples:Oranges:Pears = 2:3:5. If oranges and pears total 120, how many apples are there?

Working:

Answer: _____ **(3 marks)**

Q18. [H] [WP] Cross-Thread

Mrs. Tan spent $25 on groceries, then deposited 1/4 of the remainder. If $45 was left, how much did she start with?

Working:

Answer: _____ **(3 marks)**

Q19. [M] [WP] Part-Whole

Boys:Girls = 5:3. There are 24 more boys than girls. How many children in total?

Working:

Answer: _____ **(3 marks)**

Q20. [H] [WP] Cross-Thread

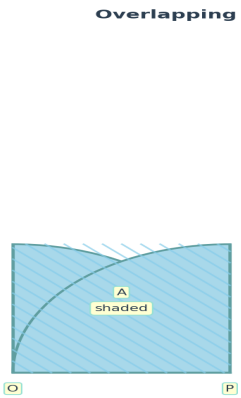
A library had 450 books. They received 120 new books, then donated 1/5 of the total. How many books remained?

Working:

Answer: _____ **(3 marks)**

Q21. [H] [G] G5-Composite Overlap

The figure shows two overlapping quarter circles with radius 10 cm. The shaded area OBC is 30 cm². (a) Find the area of figure OABCD. (b) Find the perimeter of figure OABCD.

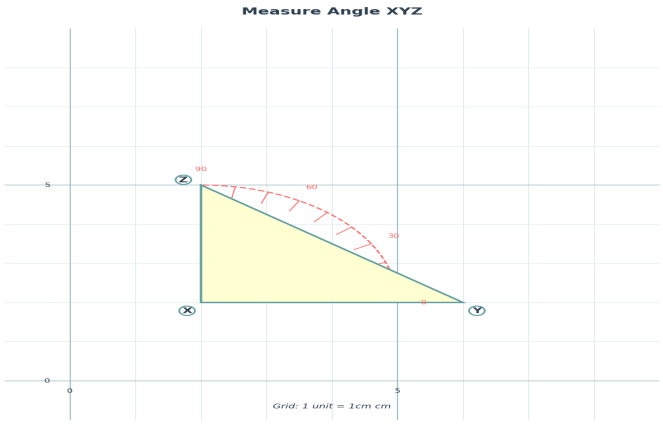


Working:

Answer: _____ **(5 marks)**

Q22. [H] [G] G6-Grid Construction

Triangle ABC is drawn on the grid. (a) Measure angle ACB. (b) Complete trapezium BCDE where BC is parallel to DE, AB = BE, and DE is twice as long as BC.

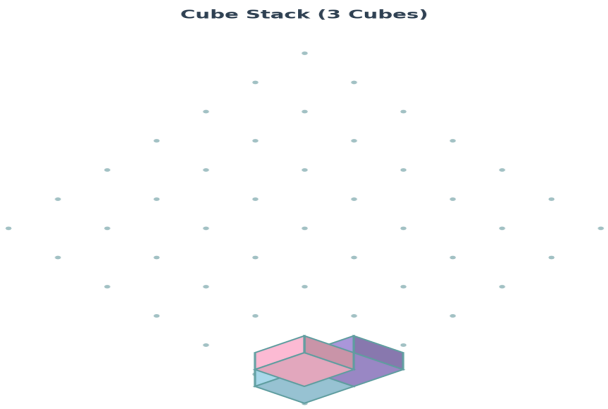


Working:

Answer: _____ **(4 marks)**

Q23. [H] [G] G7-3D Visualization

The solid is made up of 8 cubes. (a) Draw the front view of the solid on the grid. (b) What is the maximum number of cubes that can be added without changing the front and side views?

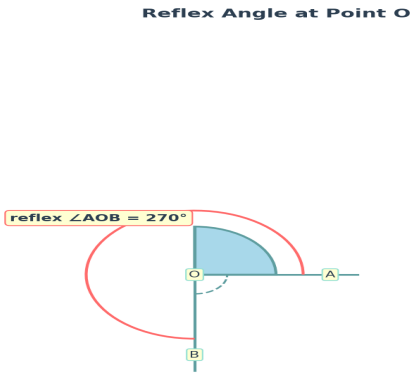


Working:

Answer: _____ **(4 marks)**

Q24. [H] [G] G8-Angle Chasing

In the figure, ABCD is a rhombus and ADEF is a trapezium with AF parallel to DE. Given angle DFE = 21°, angle BCD = 108°, and angle DAB = 33°, find angles x and y.

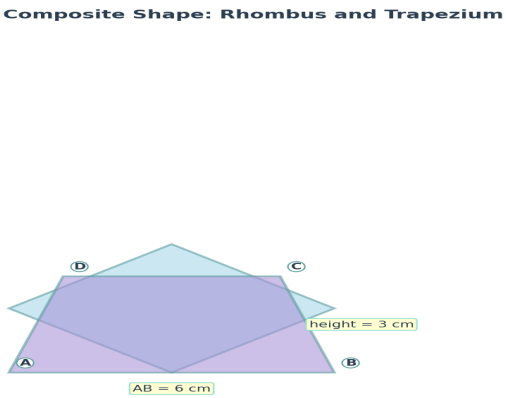


Working:

Answer: _____ **(5 marks)**

Q25. [M] [G] G5-Composite Overlap

Five squares are drawn inside a rectangle. Square X has side 4 cm. Find (a) the side of square Y, (b) the length of the rectangle, (c) how many 2cm cubes can be made from the five squares.



Working:

Answer: _____ **(5 marks)**

Q26. [E] [G] G1-Angle Reasoning

In the diagram, AOB is a straight line. If angle AOC = 65°, find angle COB.

Working:

Answer: _____ **(2 marks)**

Q27. [M] [G] G8-Angle Chasing

Three angles meet at a point. Two angles are 120° and 85°. Find the third angle.

Working:

Answer: _____ **(2 marks)**

Q28. [M] [G] G1-Angle Reasoning

Two straight lines intersect. If one angle is 40°, find the other three angles.

Working:

Answer: _____ **(2 marks)**

Q29. [M] [G] G2-Area

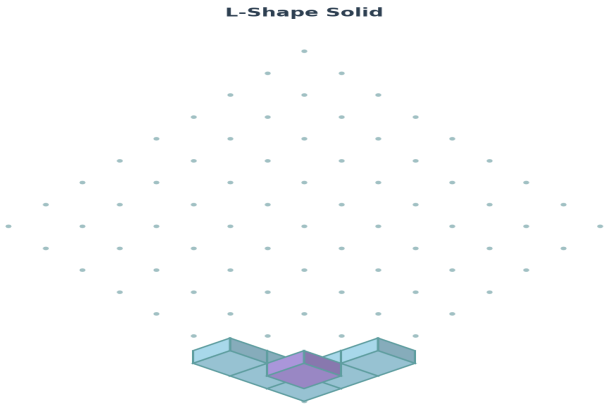
In triangle ABC, angle A = 50° and angle B = 70°. Find angle C.

Working:

Answer: _____ **(2 marks)**

Q30. [M] [G] G2-Composite

The figure shows an L-shape made of two rectangles. Find its area and perimeter.



Working:

Answer: _____ **(3 marks)**

Q31. [M] [G] G2-Area

A rectangular sheet has a smaller rectangle cut out. Find the remaining area.

Working:

Answer: _____ **(3 marks)**

Q32. [M] [G] G3-Volume

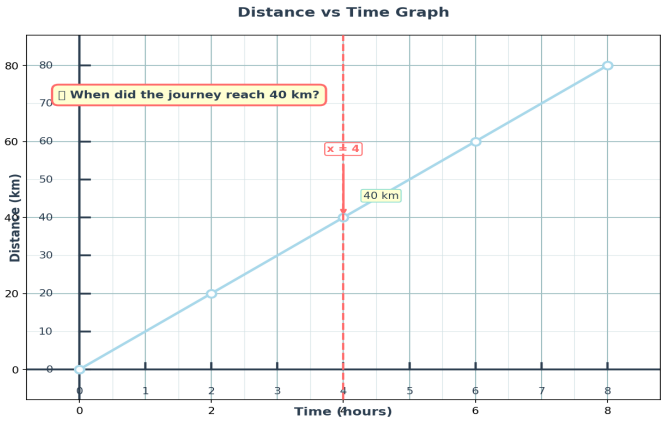
Find the volume of a cuboid with length 6 cm, breadth 4 cm, and height 3 cm.

Working:

Answer: _____ **(2 marks)**

Q33. [H] [DI] DI1-Line Graph

The line graph shows T-shirts left unsold over 7 days. (a) Which day had the most sales? (b) What percentage were sold in the first 5 days? (c) On Day 3, the shop collected $192 after 20% discount. What was the original price per T-shirt?

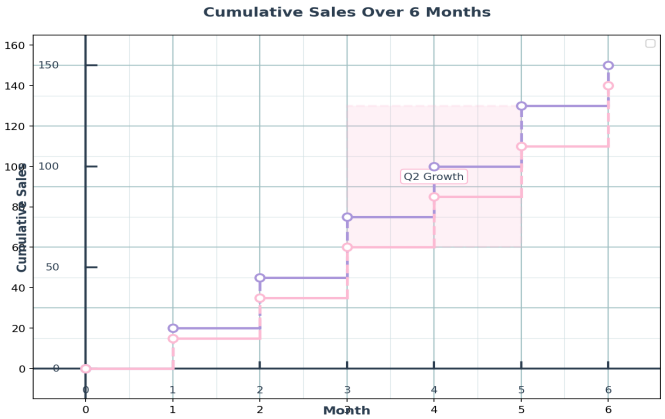


Working:

Answer: _____ **(6 marks)**

Q34. [M] [DI] DI1-Line Graph

Find the total number of T-shirts sold in the first 3 days and the range of daily sales.



Working:

Answer: _____ **(3 marks)**

Q35. [M] [DI] DI2-Bar Graph

The bar chart shows books read by 5 students. (a) Who read the most books? (b) Calculate the average. (c) What percentage of total books were read by Ali and Ben together?

Working:

Answer: _____ **(5 marks)**

Q36. [E] [DI] DI2-Bar Graph

Find the total number of books read and the range of the data.

Working:

Answer: _____ **(2 marks)**

Q37. [M] [DI] DI3-Pie Chart

The pie chart shows Isabelle's spending. Transport is 24% and equals 2/5 of food spending. Savings is 25% of transport. (a) What percentage is food? (b) If savings is $90, what is total spending?

Working:

Answer: _____ **(5 marks)**

Q38. [H] [DI] DI3-Pie Chart

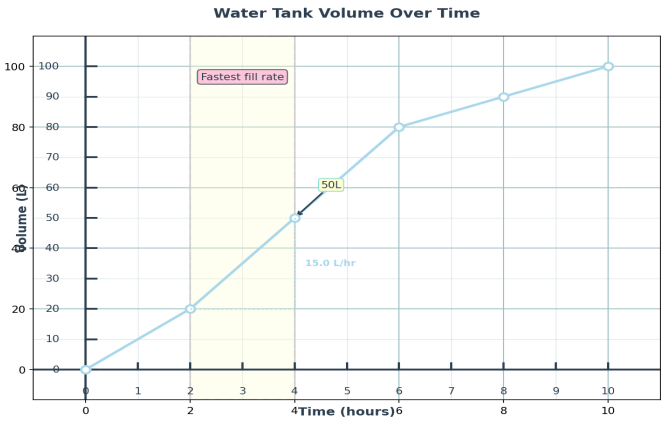
The sector for apples is 72° and represents 60 apples. Find the total number of fruits.

Working:

Answer: _____ **(3 marks)**

Q39. [M] [DI] DI1-Line Graph

Calculate the average temperature for the week and identify days above average.



Working:

Answer: _____ **(3 marks)**

Q40. [H] [DI] DI1-Line Graph

Using the temperature data, estimate the temperature on Tuesday afternoon and explain your reasoning.

Working:

Answer: _____ **(3 marks)**

Answer Key - For Markers Only

Q 1: $480
Q 2: 90 litres
Q 3: 27 red balls
Q 4: 24 years old
Q 5: 48 pens
Q 6: $144
Q 7: 90 cupcakes
Q 8: 36 stamps
Q 9: 25 boys
Q10: 37 passengers
Q11: $160
Q12: 140 litres
Q13: 50 boys
Q14: 30 apples
Q15: 34 stickers
Q16: $200
Q17: 30 apples
Q18: $85
Q19: 96 children
Q20: 456 books
Q21: (a) 127 cm^2 (b) 45.4 cm
Q22: (a) 36° (b) See diagram
Q23: (a) See diagram (b) 8 cubes
Q24: x = 18°, y = 126°
Q25: (a) 2 cm (b) 17.5 cm (c) 38 cubes
Q26: 115°
Q27: 155°
Q28: 40°, 140°, 140°
Q29: 60°
Q30: Area = 64 cm^2, Perimeter = 34 cm
Q31: See working
Q32: 72 cm^3
Q33: (a) Day 4 (b) 83% (c) $15
Q34: Total: 60, Range: 32
Q35: (a) Cal (b) 10.2 (c) 39.2%
Q36: Total: 51, Range: 9
Q37: (a) 60% (b) $1500
Q38: 300 fruits
Q39: Avg: 30.4°C
Q40: ~30°C with explanation